

Lab Session Task 1:

You have a 100-item knapsack problem with 10 constraint conditions.

$$\begin{aligned} \text{Maximize } f(\mathbf{x}) = \sum_{i=1}^{100} v_i x_i \quad \text{Subject to } \sum_{i=1}^{100} w_{ij} x_i \leq W_j, \quad j = 1, 2, \dots, 10, \\ \text{and } x_i \in \{0, 1\}, \quad i = 1, 2, \dots, 100 \end{aligned}$$

First, we create an initial feasible solution as follows: Include items one by one in the knapsack in the order of item 1, item 2, ..., item 100. If the $(k+1)$ -th item cannot be included in the knapsack, the initial feasible solution is created by including the 1st, 2nd, ..., k th items. We define the neighbors of the current solution as follows: “Neighbors are binary strings of length 100 with the Hamming distance D from the current solution”. Three values of D are examined: $D = 1, 2, 3$. A termination condition is specified as 1,000 solution evaluations.

Under these conditions, **apply a simulated annealing algorithm to this knapsack problem with various cooling schedules** (e.g., constant high temperature, constant low temperature, constant medium temperature, quick decrease of the temperature, slow decrease of the temperature).

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Solution: A 100-dimensional binary vector.

Constraint Handling:

You can use any constraint handling technique such as

(i) **Repair.** Items are removed in a pre-specified order or randomly if the solution is infeasible (until it becomes feasible). This repair process is not including in a solution evaluation count. For example, when a feasible solution 10001000...00 is obtained from a current solution 11101000...00 by removing two items, only the obtained feasible solution is counted as a new examined solution.

(ii) **Penalty.** If the solution is infeasible, a penalty value (which depends on the total constraint violation) is added to $f(\mathbf{x})$. In this case, every solution is handled as an examined solution even if it is infeasible.

The results are compared by drawing the following figure:

